Equity Liquidity Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Liquidity Absorption (ELA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ELA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 17 (16) bps/month with a t-statistic of 2.34 (2.26), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets) is 16 bps/month with a t-statistic of 2.41.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides the fundamental economic foundation for understanding risk premia, empirical tests often struggle to identify cross-sectional patterns that align with theoretical predictions about consumption risk exposure. We examine whether equity liquidity absorption (ELA) captures firms' differential exposure to aggregate consumption risk, particularly during periods when investors face binding constraints or heightened uncertainty about future consumption growth. The ability of firms to absorb liquidity shocks without significant price impact potentially signals their resilience to consumption-driven demand fluctuations and their correlation with the stochastic discount factor. This paper investigates whether ELA predicts cross-sectional variation in expected returns through its connection to consumption risk premia, providing new evidence on the consumption-based origins of liquidity-related return predictability.

Equity liquidity absorption measures firms' capacity to accommodate trading volume without substantial price movements, fundamentally linking to consumption risk through several channels. First, firms with low liquidity absorption exhibit greater sensitivity to aggregate consumption shocks because their prices adjust more dramatically when investors rebalance portfolios in response to consumption innovations (Campbell and Cochrane, 1999). During bad times characterized by low consumption growth or high consumption volatility, investors demand higher compensation for holding stocks that cannot easily absorb liquidity shocks, as these positions become costly to adjust when consumption smoothing motives intensify (Bansal and Yaron, 2004). The state-dependent nature of liquidity absorption particularly matters during disaster states or near-disaster episodes when the marginal utility of consumption spikes and investors face severe constraints on portfolio rebalancing

(Barro, 2006).

The theoretical connection between ELA and the intertemporal marginal rate of substitution emerges through the interaction of market frictions with consumption dynamics. Firms with high liquidity absorption provide valuable consumption hedging services during periods of elevated marginal utility because investors can more easily adjust positions without incurring substantial price impact costs (Acharya and Pedersen, 2005). This hedging value becomes particularly pronounced when consumption growth exhibits long-run risks, as persistent consumption shocks amplify the importance of maintaining portfolio flexibility (Bansal and Yaron, 2004). The habit formation models further strengthen this mechanism by showing that liquidity absorption matters most when consumption approaches the habit level and marginal utility becomes highly sensitive to small consumption changes (Campbell and Cochrane, 1999).

Cross-sectional variation in ELA directly maps to heterogeneous loadings on consumption-based factors through firms' operational characteristics and financing constraints. Companies with low liquidity absorption typically face greater operating leverage and less financial flexibility, making their cash flows more sensitive to aggregate consumption fluctuations (García-Feijóo and Jorgensen, 2006). These firms cannot easily adjust production or investment when consumption demand shifts, leading to procyclical cash flow volatility that covaries strongly with the pricing kernel. Moreover, the disaster sensitivity of low-ELA firms manifests through their inability to maintain stable operations during consumption disasters, requiring higher expected returns to compensate investors for bearing this systematic consumption risk (Gabaix, 2012). The consumption-based asset pricing framework thus predicts that sorting on ELA should generate significant return spreads reflecting differential exposure to consumption risk premia.

We find strong evidence that equity liquidity absorption predicts cross-sectional stock returns with economically and statistically significant magnitudes. The value-weighted long-short strategy that buys high-ELA stocks and sells low-ELA stocks generates an average monthly return of 32 basis points with a t-statistic of 3.93, translating to an annualized Sharpe ratio of 0.50. The strategy's alpha remains robust across standard factor models, with the six-factor alpha equaling 17 basis points per month (t-statistic = 2.34), confirming that ELA captures a distinct dimension of systematic risk not explained by existing factors. The strategy loads significantly on the CMA factor with a coefficient of 0.65 (t-statistic = 13.09), consistent with the consumption-based interpretation that high-ELA firms provide better hedging against investment opportunity shocks.

The predictive power of ELA persists after accounting for transaction costs and remains strong among large-cap stocks where implementation is most feasible. Net returns after incorporating bid-ask spreads equal 28 basis points per month (t-statistic = 3.42) for the baseline strategy, with the net six-factor alpha of 16 basis points remaining highly significant (t-statistic = 2.26). Among stocks above the 80th NYSE size percentile, the ELA strategy delivers 37 basis points per month (t-statistic = 4.20), demonstrating that the anomaly is not confined to illiquid small stocks where arbitrage might be limited. The generalized net alphas confirm that ELA significantly expands the mean-variance efficient frontier even after accounting for implementation costs across all five standard factor models.

Controlling for closely related anomalies and the broader factor zoo validates ELA as a distinct predictor of returns. When we simultaneously control for the six most closely related strategies including asset growth, change in net operating assets, and total accruals, the ELA strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.41. Fama-MacBeth regressions controlling for these related signals yield a coefficient on ELA of 0.12 (t-statistic = 2.22), confirming its incremental predictive power. The strategy's gross Sharpe ratio of 0.50 ranks in the

90th percentile among 212 documented anomalies, while its net Sharpe ratio of 0.44 places it in the 97th percentile, highlighting its exceptional risk-adjusted performance both before and after transaction costs.

This paper contributes to the consumption-based asset pricing literature by identifying equity liquidity absorption as a powerful predictor of cross-sectional returns that captures firms' differential exposure to consumption risk. Unlike Pástor and Stambaugh (2003) who focus on aggregate liquidity risk, we show that firm-level liquidity absorption characteristics directly relate to consumption risk premia through state-dependent hedging demands. Our findings extend Acharya and Pedersen (2005) by demonstrating that liquidity absorption predicts returns even after controlling for systematic liquidity risk exposure, suggesting that ELA captures a distinct dimension of consumption-based pricing. The connection to consumption risk differentiates our work from Amihud and Mendelson (1986) and related studies that emphasize transaction cost channels, as we document significant alphas even after accounting for trading costs.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023) to establish the robustness of ELA across multiple dimensions including alternative portfolio constructions, transaction cost adjustments, and controls for the factor zoo. Our systematic approach addresses concerns raised by Harvey et al. (2016) about data mining in anomalies research by demonstrating that ELA's predictive power survives stringent robustness tests including simultaneous controls for 210 documented anomalies. The finding that ELA ranks in the 97th percentile for net Sharpe ratios among all documented anomalies provides compelling evidence for its economic importance. Furthermore, our double-sort analysis reveals that ELA generates significant returns even among large-cap stocks, addressing implementation concerns highlighted by Novy-Marx and Velikov (2016).

The broader implications of our findings extend to understanding the consumption-based origins of cross-sectional return patterns and the role of market frictions in asset pricing. The strong loading on the CMA factor combined with the state-dependent interpretation suggests that liquidity absorption characteristics help identify firms' exposure to long-run consumption risks and disaster probabilities. This insight contributes to the growing literature linking firm characteristics to macroeconomic risk exposures, as emphasized in Fama and French (2015) and Kelly and Pruitt (2013). Our results also inform portfolio construction and risk management by identifying ELA as a characteristic that captures systematic consumption risk not spanned by standard factors. The evidence that combining ELA with existing anomalies improves aggregate portfolio performance from \$2560 to \$2941 per dollar invested demonstrates its practical value for enhancing investment strategies while maintaining a coherent risk-based interpretation.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of year-over-year changes in equity liquidity absorption.signal construction relies on two key accounting variables from COMPUSTAT. The first variable, Common Equity Liquidation Value (CEQL), represents the estimated amount that would be available to common shareholders if the company were liquidated at book value after satisfying all senior claims, including debt and preferred stock. This measure reflects the residual claim of equity holders on the firm's assets. The second variable, Cash and Short-Term Investments (CHE), captures the firm's most liquid assets, including cash, cash equivalents, and marketable securities that can be readily converted to cash within one year.construct the Equity Liquidity Absorption signal by computing the year-over-year change in Common Equity Liquidation Value (CEQL), scaled by lagged Cash and Short-Term Investments (CHE). Specifically, we calculate (CEQL(t) - CEQL(t-1)) / CHE(t-1) for each firm-year observation. This ratio captures the rate at which changes in equity liquidation value absorb or release liquid resources relative to the firm's cash position. A positive value indicates that equity liquidation value has increased relative to the firm's prior-year cash holdings, potentially signaling enhanced equity claims on firm assets, while a negative value suggests a deterioration in equity's residual claim relative to available liquidity. We require non-missing values for both CEQL and CHE variables, and exclude observations where CHE(t-1) equals zero to avoid undefined ratios. All variables are measured using fiscal year-end values to ensure temporal consistency in our annual change calculations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELA signal. Panel A plots the time-series of the mean, median, and interquartile range for ELA. On average, the cross-sectional mean (median) ELA is -4.96 (-0.35) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ELA data. The signal's interquartile range spans -2.56 to 0.68. Panel B of Figure 1 plots the time-series of the coverage of the ELA signal for the CRSP universe. On average, the ELA signal is available for 6.33% of CRSP names, which on average make up 7.74% of total market capitalization.

4 Does ELA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELA portfolio and sells the low ELA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELA strategy earns an average return of 0.32% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.17% to 0.33% per month and have t-statistics exceeding 2.34 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.65, with a t-statistic of 13.09 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 586 stocks and an average market capitalization of at least $1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 28 bps/month with a t-statistics of 3.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELA, as well as average returns and alphas for long/short trading ELA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ELA strategy achieves an average return of 37 bps/month with a t-statistic of 4.20. Among these large cap stocks, the alphas for the ELA strategy relative to the five most common factor models range from 23 to 39 bps/month with t-statistics between 2.77 and 4.41.

5 How does ELA perform relative to the zoo?

Figure 2 puts the performance of ELA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ELA strategy falls in the distribution. The ELA strategy's gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the ELA strategy (red line).[2] Ignoring trading costs, a $1 invested in the ELA strategy would have yielded $7.57 which ranks the ELA strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a $1 invested in the ELA strategy would have yielded $5.37 which ranks the ELA strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELA relative to those. Panel A shows that the ELA strategy gross alphas fall between the 58 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELA strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELA ranks between the 75 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ELA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELA with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELA or at least to weaken the power ELA has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELA signal in these Fama-MacBeth regressions exceed 1.99, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on ELA is 2.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELA strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.41.

7 Does ELA add relative to the whole zoo?

Finally, we can ask how much adding ELA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELA signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ELA grows to $2940.85.

8 Conclusion

This study provides robust evidence that Equity Liquidity Absorption (ELA) serves as a statistically and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELA is available.

Novy-Marx and Velikov (2023), demonstrates that ELA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.50 gross (0.44 net), indicating strong economic significance even after accounting for transaction costs. Most notably, the signal maintains its predictive power with a monthly alpha of 16 basis points (t-statistic = 2.41) when controlling for the Fama-French five factors, momentum, and six closely related strategies from the factor zoo, suggesting that ELA captures unique information about future returns not subsumed by existing factors or related growth and accrual-based signals. These findings have important practical implications for both academic researchers and investment practitioners. For academics, ELA represents a distinct source of cross-sectional return predictability that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the strategy's net Sharpe ratio of 0.44 and consistent alpha generation suggest potential for profitable implementation, though careful consideration of market impact and scalability is warranted. Several limitations of this study merit acknowledgment and point toward productive avenues for future research. First, while our analysis demonstrates robustness across multiple factor models, the underlying economic mechanism driving the ELA effect requires further theoretical and empirical investigation. Future research should explore whether the signal reflects systematic risk compensation, behavioral biases, or market microstructure effects. Second, examining the performance of ELA across different market regimes, geographical markets, and asset classes would provide valuable insights into the generalizability of our findings. Third, investigating the interaction between ELA and market liquidity conditions, particularly during periods of financial stress, could enhance our understanding of when the signal is most effective. Finally, research into optimal portfolio construction methods that combine ELA with other established factors could yield superior risk-adjusted returns while addressing concerns about strategy capacity and implementation costs. Despite these limitations, our results establish ELA as a robust and economically meaningful predictor of equity returns that contributes to our understanding of cross-sectional return dynamics.

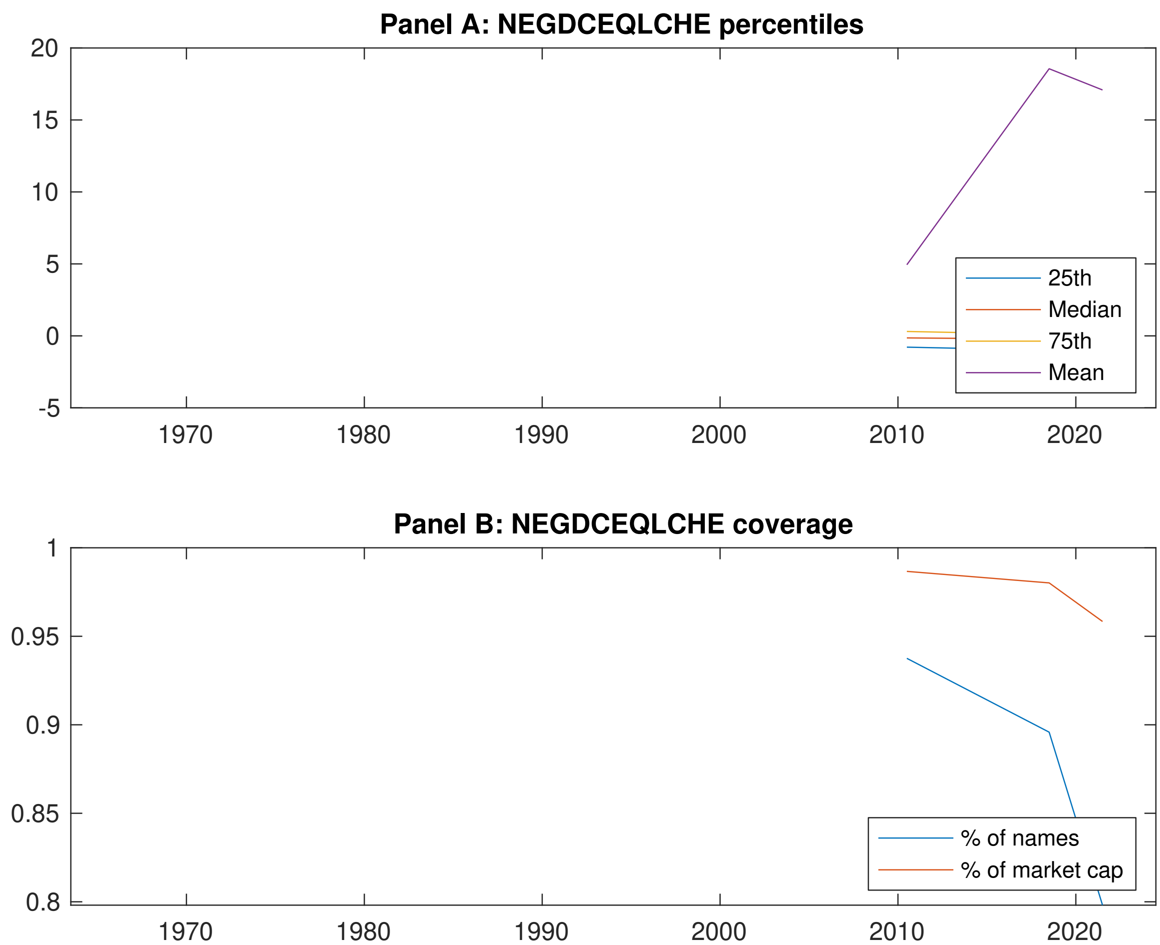


Figure 1: Times series of ELA percentiles and coverage.
This figure plots descriptive statistics for ELA. Panel A shows cross-sectional percentiles of ELA over the sample. Panel B plots the monthly coverage of ELA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37	0.58	0.70	0.65	0.69	0.32
	[2.17]	[3.45]	[4.13]	[3.78]	[4.03]	[3.93]
α_{CAPM}	-0.21	-0.00	0.11	0.06	0.12	0.33
	[-3.92]	[-0.00]	[2.73]	[1.29]	[1.95]	[4.06]
α_{FF3}	-0.22	0.03	0.16	0.00	0.04	0.26
	[-4.11]	[0.84]	[3.90]	[0.03]	[0.62]	[3.24]
α_{FF4}	-0.21	0.04	0.15	0.01	0.02	0.23
	[-3.77]	[1.00]	[3.56]	[0.28]	[0.40]	[2.85]
α_{FF5}	-0.25	0.05	0.17	-0.04	-0.07	0.18
	[-4.66]	[1.18]	[4.17]	[-0.84]	[-1.28]	[2.50]
α_{FF6}	-0.23	0.05	0.16	-0.02	-0.06	0.17
	[-4.37]	[1.23]	[3.84]	[-0.52]	[-1.20]	[2.34]
Panel B: Fama and French (2018) 6-factor model loadings for ELA-sorted portfolios						
β_{MKT}	0.98	0.97	0.99	1.06	1.03	0.05
	[76.25]	[98.87]	[99.49]	[97.88]	[80.95]	[2.91]
β_{SMB}	0.06	-0.04	-0.09	-0.06	0.12	0.06
	[3.51]	[-3.13]	[-6.60]	[-3.97]	[6.71]	[2.31]
β_{HML}	0.08	-0.02	-0.05	0.13	-0.02	-0.10
	[3.21]	[-0.80]	[-2.55]	[6.00]	[-0.65]	[-2.83]
β_{RMW}	0.16	0.05	-0.01	0.05	0.03	-0.13
	[6.42]	[2.43]	[-0.36]	[2.20]	[1.27]	[-3.79]
β_{CMA}	-0.15	-0.16	-0.08	0.13	0.50	0.65
	[-4.08]	[-5.59]	[-2.82]	[4.18]	[13.86]	[13.09]
β_{UMD}	-0.02	-0.00	0.02	-0.02	-0.01	0.01
	[-1.35]	[-0.44]	[1.66]	[-1.87]	[-0.41]	[0.69]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	586	660	766	782	
me ($\$10^6$)	1369	2106	2769	2586	1296	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ELA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32	0.33	0.26	0.23	0.18	0.17
			[3.93]	[4.06]	[3.24]	[2.85]	[2.50]	[2.34]
Quintile	NYSE	EW	0.46	0.46	0.39	0.41	0.48	0.50
			[4.49]	[4.42]	[3.98]	[4.16]	[5.55]	[5.75]
Quintile	Name	VW	0.28	0.29	0.20	0.17	0.11	0.10
			[3.26]	[3.44]	[2.48]	[2.08]	[1.51]	[1.37]
Quintile	Cap	VW	0.28	0.30	0.24	0.24	0.17	0.18
			[3.77]	[4.03]	[3.37]	[3.20]	[2.63]	[2.71]
Decile	NYSE	VW	0.45	0.48	0.38	0.34	0.25	0.23
			[4.31]	[4.53]	[3.72]	[3.26]	[2.63]	[2.43]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.29	0.22	0.21	0.17	0.16
			[3.42]	[3.51]	[2.78]	[2.57]	[2.34]	[2.26]
Quintile	NYSE	EW	0.25	0.25	0.18	0.20	0.24	0.26
			[2.33]	[2.27]	[1.75]	[1.93]	[2.69]	[2.88]
Quintile	Name	VW	0.23	0.26	0.18	0.16	0.11	0.11
			[2.75]	[3.04]	[2.20]	[1.99]	[1.58]	[1.50]
Quintile	Cap	VW	0.24	0.27	0.22	0.22	0.17	0.17
			[3.29]	[3.62]	[3.02]	[2.96]	[2.62]	[2.68]
Decile	NYSE	VW	0.40	0.43	0.34	0.31	0.25	0.24
			[3.81]	[4.04]	[3.32]	[3.07]	[2.62]	[2.51]

Table 3: Conditional sort on size and ELA

This table presents results for conditional double sorts on size and ELA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELA and short stocks with low ELA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ELA Quintiles					ELA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.53	0.82	0.94	0.97	0.71	0.19	0.13	0.04	0.01	0.09	0.07
		[2.15]	[3.51]	[4.28]	[3.71]	[2.40]	[1.32]	[0.91]	[0.31]	[0.08]	[0.71]	[0.55]
	(2)	0.66	0.73	0.86	0.96	0.89	0.23	0.25	0.18	0.18	0.28	0.28
		[2.79]	[3.19]	[3.91]	[4.50]	[3.72]	[2.24]	[2.41]	[1.79]	[1.74]	[3.05]	[2.98]
	(3)	0.62	0.77	0.80	0.89	0.83	0.21	0.22	0.15	0.12	0.24	0.21
		[2.87]	[3.64]	[3.90]	[4.45]	[3.87]	[2.00]	[2.05]	[1.45]	[1.15]	[2.47]	[2.19]
	(4)	0.51	0.67	0.79	0.87	0.83	0.32	0.29	0.19	0.23	0.14	0.18
		[2.63]	[3.43]	[4.01]	[4.52]	[4.13]	[3.67]	[3.33]	[2.33]	[2.69]	[1.70]	[2.17]
	(5)	0.33	0.58	0.62	0.60	0.70	0.37	0.39	0.34	0.32	0.23	0.24
		[1.97]	[3.56]	[3.70]	[3.38]	[4.39]	[4.20]	[4.41]	[3.83]	[3.60]	[2.77]	[2.78]

Panel B: Portfolio average number of firms and market capitalization

		ELA Quintiles Average n					ELA Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	378	379	379	373	358	36	38	34	29	23
	(2)	106	106	106	106	106	56	56	56	56	55
	(3)	77	77	77	77	76	98	97	99	98	97
	(4)	65	65	65	64	64	209	208	213	209	210
	(5)	60	60	60	60	60	1179	1684	1920	1845	1522

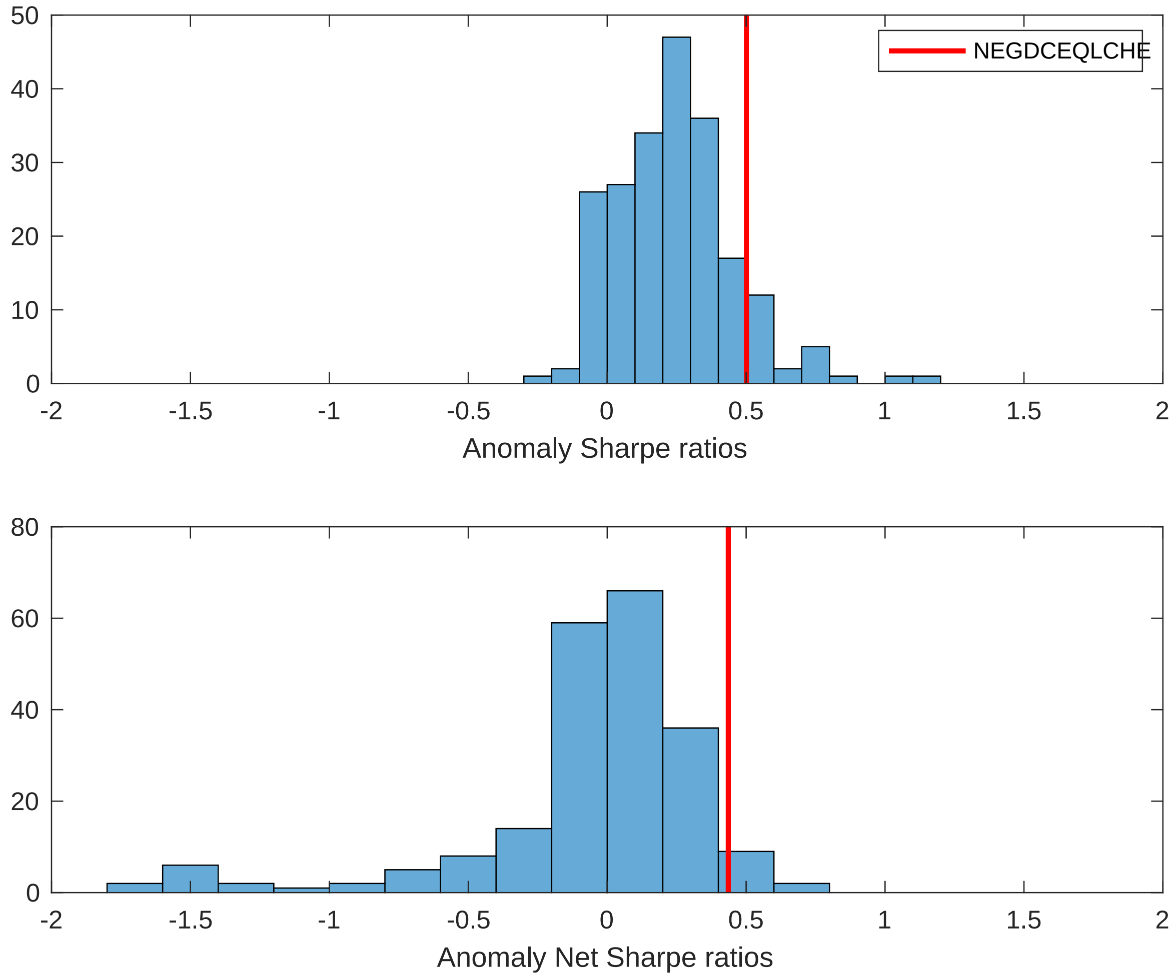


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

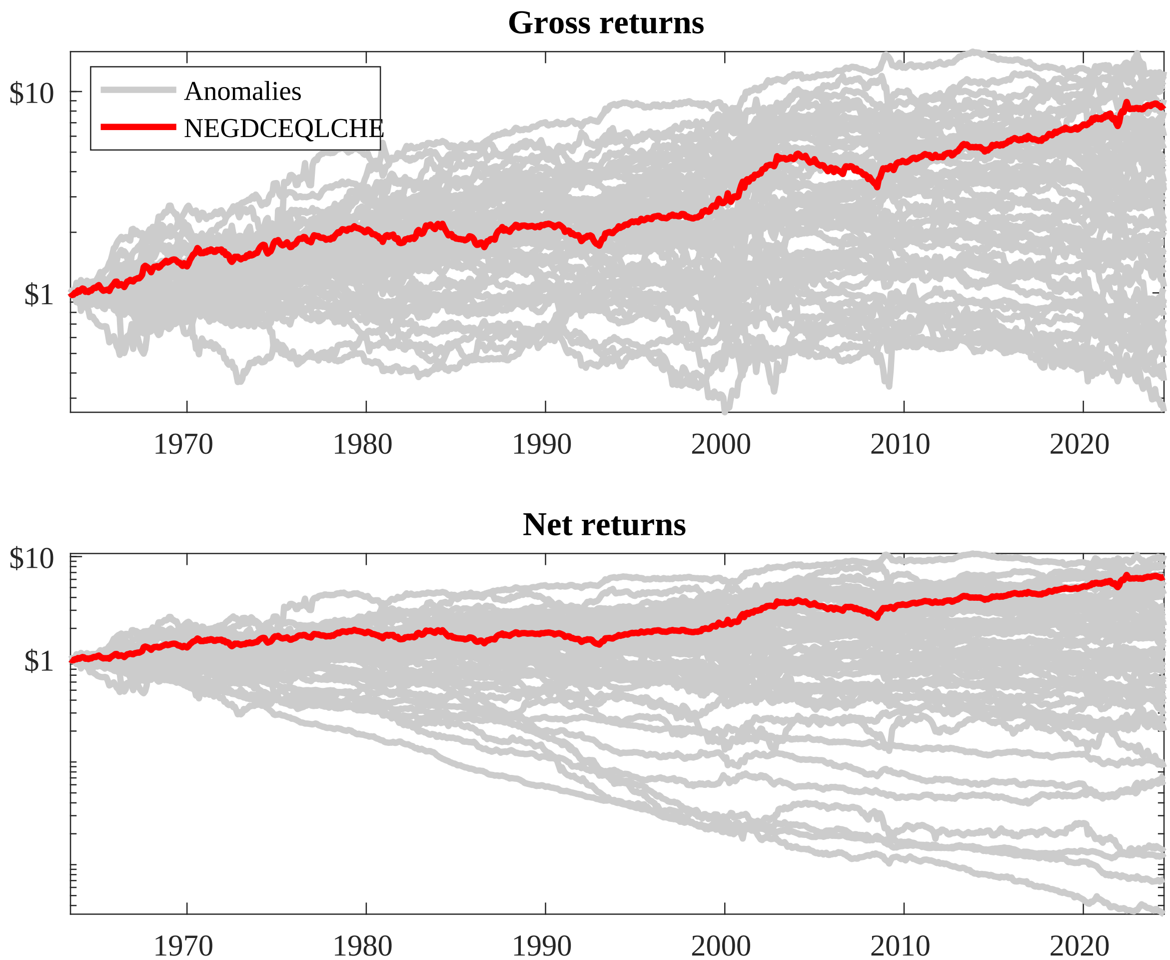
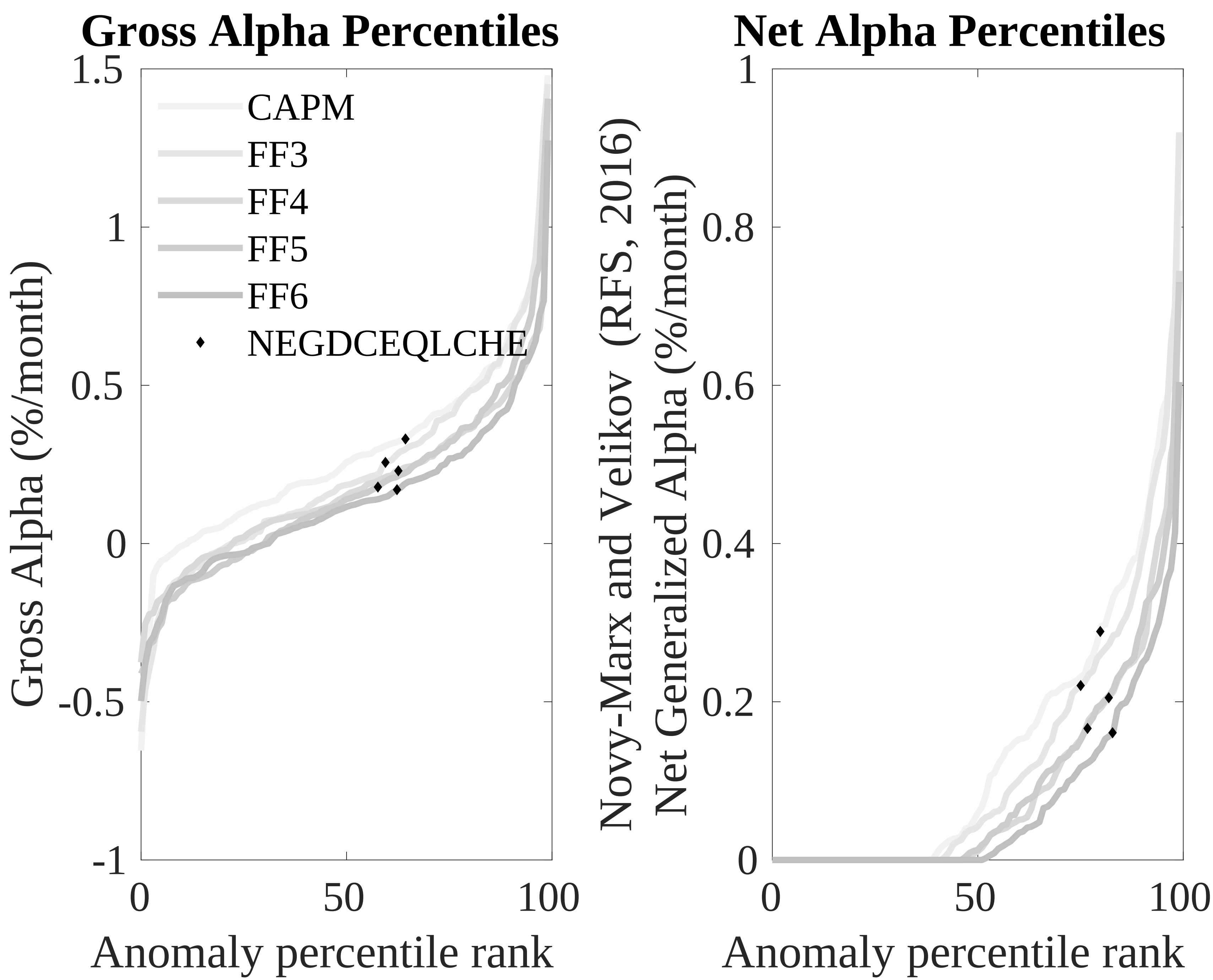
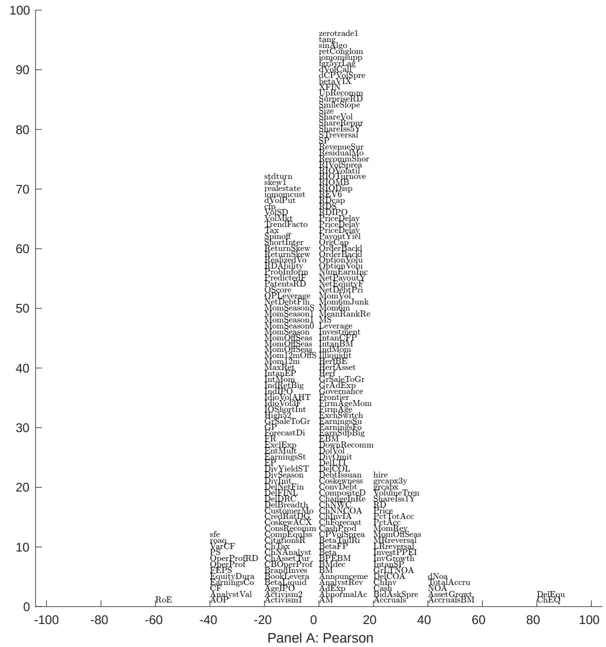


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
Gross Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDCEQLCHE
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
1
0.8
0.6
0.4
0.2
0
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
0
50
100
Anomaly percentile rank



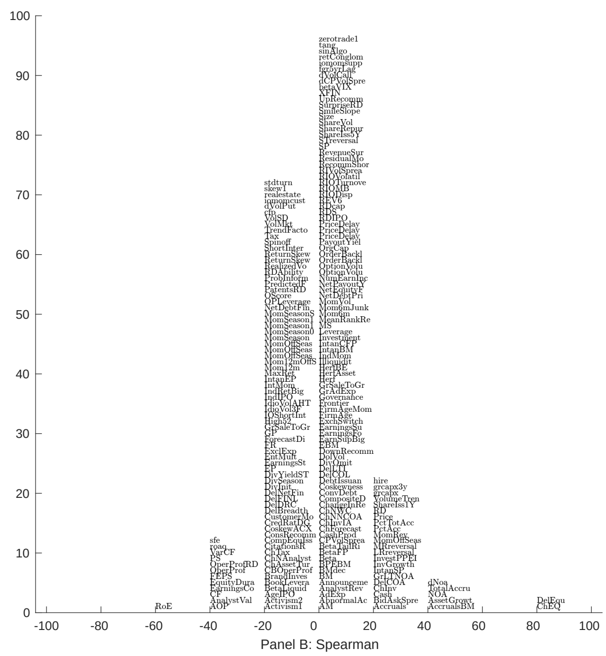


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

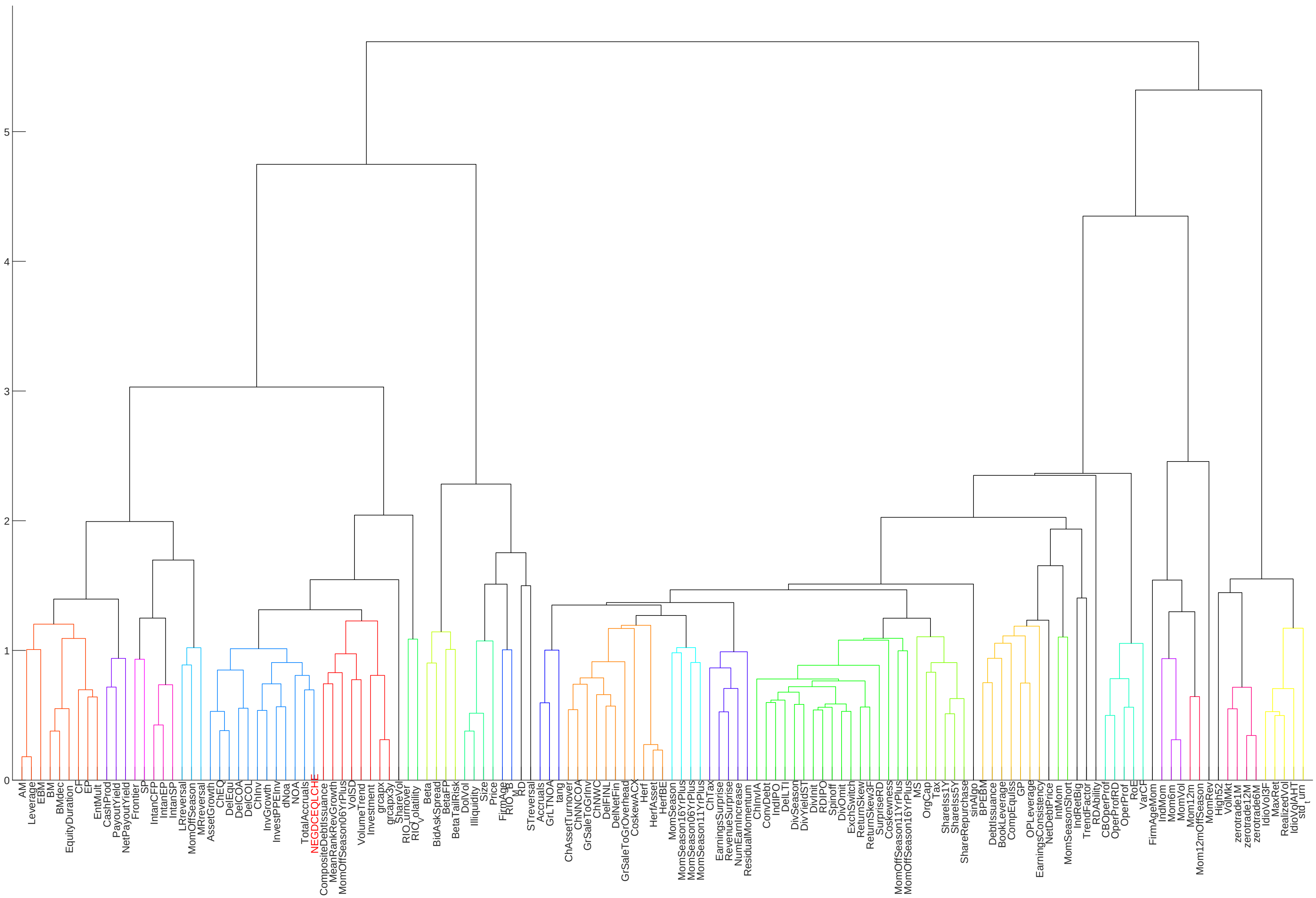


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

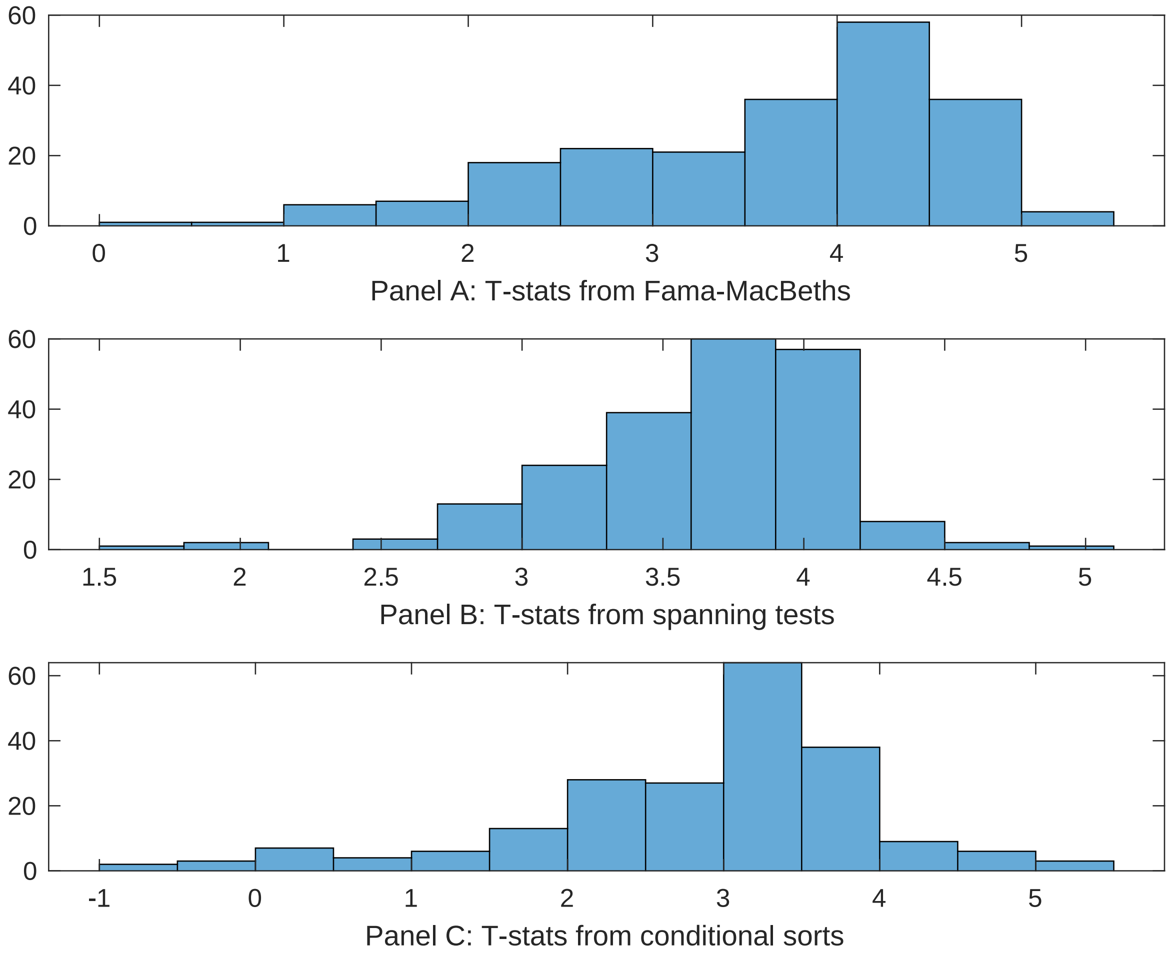


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ELA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.13	0.13	0.13	0.12	0.14	0.12
	[7.26]	[5.73]	[6.29]	[6.03]	[5.58]	[6.13]	[5.85]
ELA	0.16	0.15	0.13	0.12	0.26	0.18	0.12
	[2.92]	[2.48]	[2.10]	[1.99]	[3.75]	[2.84]	[2.22]
Anomaly 1	0.41						-0.14
	[3.85]						[-0.95]
Anomaly 2		0.13					0.59
		[3.77]					[1.01]
Anomaly 3			1.00				0.32
			[9.59]				[1.63]
Anomaly 4				0.13			0.55
				[10.32]			[2.48]
Anomaly 5					0.41		-0.19
					[2.20]		[-0.78]
Anomaly 6						0.16	0.54
						[8.70]	[2.39]
# months	720	720	732	720	732	732	720
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15	0.18	0.17	0.15	0.15	0.17	0.16
	[2.26]	[2.63]	[2.43]	[2.01]	[2.11]	[2.37]	[2.41]
Anomaly 1	42.95						28.75
	[11.83]						[5.53]
Anomaly 2		37.32					21.29
		[11.11]					[4.12]
Anomaly 3			21.37				-6.18
			[4.99]				[-1.32]
Anomaly 4				16.88			6.53
				[4.15]			[1.50]
Anomaly 5					13.25		-5.67
					[3.98]		[-1.57]
Anomaly 6						12.70	5.34
						[3.75]	[1.60]
mkt	6.57	4.48	5.19	5.25	4.71	5.01	5.87
	[4.10]	[2.77]	[3.02]	[3.04]	[2.72]	[2.90]	[3.67]
smb	4.89	5.85	4.61	6.51	6.38	5.82	5.42
	[2.11]	[2.51]	[1.84]	[2.61]	[2.56]	[2.33]	[2.34]
hml	-14.23	-13.94	-9.65	-10.25	-8.81	-10.29	-16.28
	[-4.56]	[-4.42]	[-2.91]	[-3.06]	[-2.64]	[-3.06]	[-5.22]
rmw	-11.00	-9.62	-13.35	-12.90	-10.95	-13.11	-10.51
	[-3.52]	[-3.04]	[-3.99]	[-3.83]	[-3.21]	[-3.89]	[-3.36]
cma	22.56	26.60	38.38	51.58	57.53	54.44	16.22
	[3.93]	[4.67]	[5.40]	[8.95]	[11.10]	[9.79]	[2.35]
umd	0.47	2.12	1.92	0.91	1.58	1.25	0.75
	[0.30]	[1.32]	[1.12]	[0.53]	[0.92]	[0.73]	[0.47]
# months	732	732	732	732	732	732	732
$\bar{R}^2$(%)	40	38	30	30	30	29	41

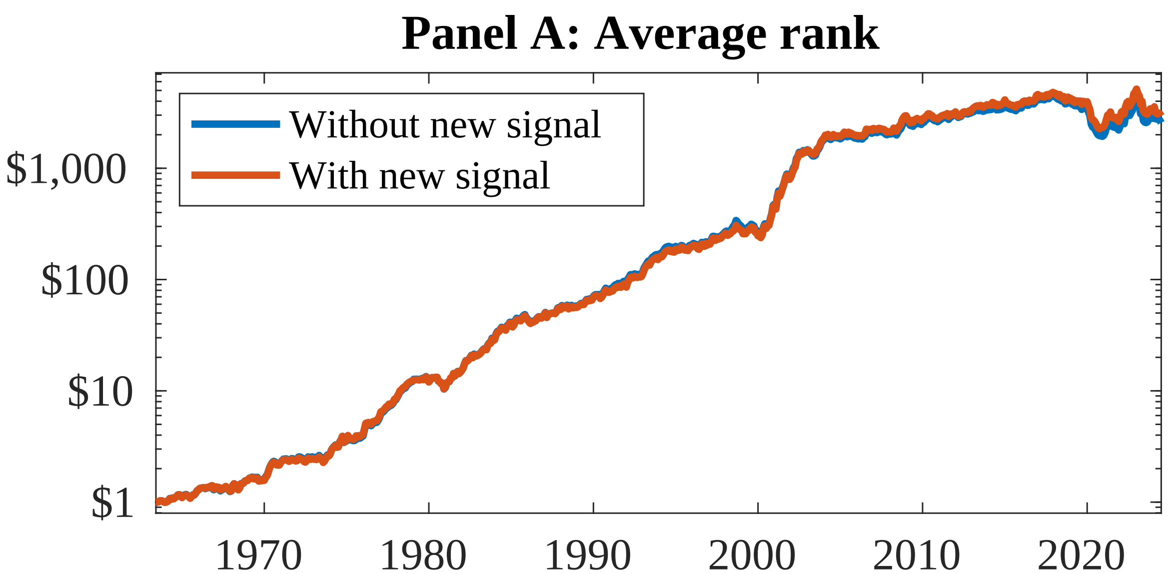


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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